

Avery Bettesworth

Edmonton, AB | 403-848-1760 | abettesw@ualberta.ca

GitHub:// Betts6430 ▪ LinkedIn:// avery-bettesworth ▪ Devpost:// abettesw



Engineering
Co-op Program

ACADEMIC & CO-OP STATUS

Computer Engineering, BSc Co-op, University of Alberta	Class of 2028
Completed Academic Terms	5 of 8
Completed Co-op Work Terms	2 of 5
Availability Starting May 2026	4 – 8 months

TECHNICAL SKILLS

Programming Languages:	Python, Java, C, C++, MATLAB, Bash, ARM/MIPS assembly
Libraries and Databases:	NumPy, Pillow, SciPy, Selenium WebDriver, Pygame, BeautifulSoup, SQL, MongoDB
Hardwares and Tools:	Microsoft Office (Excel, Word, PowerPoint, Access), Git, Arduino, RaspberryPi, 3D CAD, Linux, LTSpice

WORK EXPERIENCE

Computer Technician | Alberta Computers for Schools | Edmonton, AB Jan – May 2026

- Collaborated closely within a 6 person team to execute full end-to-end refurbishment on computer systems, performing hardware diagnosis, component replacement, BIOS optimization, OS deployment, and QA validation.
- Assembled and tested 200+ enterprise devices (\$75,000+ in equipment), creating detailed technical reports and satisfying client specifications.
- Identified a data entry bottleneck in the computer refurbishment process and developed a Python automation tool using Selenium WebDriver and tkinter, reducing computer database logging time by 80% (5 min to 1 min) and eliminating ~3000 manual data entries per quarter.
- Implemented secure data sanitization protocols on HDDs and SSDs, ensuring privacy standards are met and data integrity is maintained.

Sales Associate | TJX Canada | Red Deer, AB Sep 2022 – Feb 2024

- Excelled in delivering customer service under high-pressure conditions, processing up to 300 transactions each day.
- Collaborated with warehouse team to process and organize incoming stock, reducing unpacking time through efficient workflows.
- Trained 3 new hires in cash register operations and warehouse duties, improving team productivity and cohesion.

PROJECTS

Terminal E-commerce | Python, SQLite | Introduction to File and Database Management CMPUT 291 Oct – Nov 2025

- Engineered a terminal-based e-commerce system using Python and SQLite, integrating a relational database to manage secure user authentication, product inventory, and transaction processing.
- Programmed complex queries to generate weekly sales metrics, past order summaries, product information, search functionalities, and more, ensuring resistance against SQL injection attacks and implementing SHA-256 password hashing.
- Designed a comprehensive tracking system that logs every user session, search query, and product view into relational tables.

FocusBoost | Python, Arduino, React | NatHACKS 2024 Nov 2024

- Led a 5-person team over a 72-hour hackathon to design and implement a real-time focus detection tool that provides motivational alerts and notifications when users lose focus in an Electron-based web application.
- Engineered a system to detect brainwaves using the BioAmp EXG Pill and Arduino, filtering EEG signals into the 0–50 Hz brainwave frequency range and isolating beta waves (14–40 Hz) using Butterworth bandpass filtering for accurate focus monitoring.
- Showcased a live demonstration, highlighting potential use cases in education and productivity in 5 minute presentation to a panel of judges.

BaccaratBot | Python May 2024 – Present

- Created a baccarat simulator with 15+ customizable strategies to test win percentages, longevities, and overall strategy viabilities.
- Collected data into excel spreadsheet and word document to further analyze and display predictive earnings and bankruptcy trends.
- Currently working on expanding into a full casino table game simulator to teach game rules and model profit maximizing strategies.

Recreation Centre Redesign | Introduction to Engineering Design, Communication, and Profession ENGG 103 Jan – Apr 2024

- Led a team of 6 group members to design a new recreation centre for a community of over 2000 people in Windermere, Edmonton.
- Tested the viability of engineering designs based on regulatory standards and budget, environmental, accessibility, and community impacts.
- Organized weekly meetings, assigned tasks, and oversaw project creation and presentation.

AWARDS & SCHOLARSHIPS

Jason Lang Scholarship, 2025
Nova Chemicals Engineering University Transfer Award, 2024

President's List, Red Deer Polytechnic, 2024
International Certificate, Red Deer Public Schools, 2023